

ClinicCare Manager: An Efficient Electronic Health Record (EHR) and Healthcare Management System using Spring Boot and React

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Abstract — Healthcare institutions often face challenges in managing patient data, optimizing clinic workflows, and ensuring timely reporting. Smaller clinics, in particular, may lack the resources to streamline services for both patients and healthcare providers. This work introduces a platform that automates patient data analysis and report generation to address these needs. Built using React, Spring Boot, and PostgreSQL, the application provides efficient, end-to-end service with robust data handling and fast processing. Performance metrics show efficient resource utilization, with CPU usage rarely exceeding 3% and heap memory remaining well within the 2048 MB limit. Compared to other healthcare applications, this platform offers greater scalability and responsiveness. Additionally, the framework supports secure authentication and role-based access, making it a reliable and scalable solution that meets the expanding demands of healthcare environments.

Keywords— *Healthcare, Web Application, React, Spring Boot, PostgreSQL, mHealth, Data Analysis.*

I. Introduction

The healthcare domain has a number of problems which include good methods of data storage, quick and reliable patient management and optimal use of resources. Most clinics as well as hospitals face these issues due to older systems or outdated technologies therefore increasing overhead for nurses and assistants making patient care move

in a slow and inefficient manner. This is where mHealth comes in as it provides a customizable platform for various cases that clinics come across while managing patients. Such m-health applications have proven to be effective for in the field of healthcare as they use innovative technology to help people with illnesses and to enhance their health through treatment adherence and assist physicians in the continuous assessment of the patient's health status as chronic illnesses need constant monitoring and check of the daily dose of medicine patients need to take. Within a single platform there will be the benefits so as to unload some of the manual efforts put in by all healthcare personnel by removing tasks of interpreting the data logging it and preparing the reports every time a new log is made. This enables Healthcare providers to spend their time on providing care for the patients.

Small and medium-sized clinics for example experience difficulties in managing data and analysing information efficiently. Patient health record management, hospital resource management and resulting in timely reporting has been widely seen as failing, which in the long run affects patient care while adding to the workload. Besides, the hamster wheel of not having contemporary technologies in the given type of organizations creates the motivation for the need of an automated system which could solve these problems.

The main aim of this work is to find an optimal solution for hospital management with less resources by focusing on automating analysis of patient data, auto generation of reports and a building a data driven inventory system. By integrating

modern technologies, this work will help enhance decision-making, improves efficiency, and supports better overall patient outcomes in small to medium-sized healthcare settings.

This paper is structured as follows: Section II provides a literature review of related works and researches. Section III explains the architecture, data collection and data processing. Section IV presents the results and discussion. Section V concludes the paper and suggests directions for future research.

II. Related Works

Numerous studies have been made on the effectiveness of m-health applications. A comprehensive meta-analysis involving 84 studies and over 31,000 participants explored how people accept e-health technologies like mHealth and wearable devices. The analysis revealed that people's attitudes had the strongest impact on their intention to use these technologies, with a notable effect size ($\beta = 0.6470$, $p = 0.00$) [1]. While the study also considered 23 factors that could influence various relationships, certain links, such as how easy people find the technology and how that impacts their intention to use it, were not significantly affected by these factors.

Studies have highlighted some ongoing challenges and often overlooked issues in mobile health, especially the role of smartphones in shaping the field. While smartphones have helped make mHealth more popular, they've also created a gap between commercial apps and scientific research. This gap has limited the potential benefits for global health. Focusing more on solid scientific research, rather than just apps, could help mobile health become more effective and scalable. Involving international experts and researchers may also lead to better healthcare solutions [2].

Understanding user behavior in health technology adoption requires a combination of established models and broader perspectives. A study on the extended TAM and UTAUT models found these frameworks effective in predicting the intention to use health technologies. However, it suggested that combining multiple frameworks could provide a deeper understanding of user needs, emphasizing the importance of factors like accessibility, equity, and social conditions through qualitative approaches [3].

M-health applications have been developed for various medical purposes, addressing both chronic illnesses and common conditions like diabetes. For example, a mobile application designed to detect melanoma utilized enhanced image processing techniques, improving accuracy in identifying skin conditions. The model was successfully integrated into the app, enabling real-time health monitoring [4].

Digital medicine spans telemedicine, AI, machine learning, and digital diagnostics, but lacks consistent terminology. Studies suggest that telemedicine and tele-rehabilitation are central fields, underscoring the need for a unified ontology

to improve research comparisons and patient care strategies [5]. For example, m-Health applications have positively impacted lung cancer patients' recovery, enhancing postoperative outcomes, quality of life, and physical activity. Tools like mobile apps and wearables help monitor symptoms, provide personalized exercises, and offer emotional support, resulting in reduced pain, anxiety, and better adherence to rehabilitation programs [6].

m-Health solutions show promise in improving recovery and care after lung cancer surgery. They also demonstrate effectiveness in diabetes management, as shown in a review of 27 studies with 2,887 patients. Using diabetes-specific m-Health apps led to reductions in HbA1c levels—0.54% for type 2 diabetes and 0.63% for type 1—while supporting self-care and daily management of blood glucose, medication adherence, and lifestyle changes. Although these findings are positive, further research is needed to confirm m-Health apps' clinical impact on diabetes management [7].

Additionally, hypertension is a medical condition that requires ongoing care. "Blood Pressure Assistant," a mHealth app, was used in a randomised controlled experiment with 297 patients in China to assess its effects on managing hypertension. Compared to those getting conventional care, participants who used the app over a six-month period shown noticeably higher increases in blood pressure control, hypertension awareness, and self-management behaviours. According to the results, mHealth treatments can improve the management of hypertension, particularly when it comes to dietary practices, physical activity, and medication adherence [8].

Incorporating m-Health and health-coaching interventions can play a pivotal role in enhancing self-management for patients with chronic diseases. A scoping review identified two main approaches: using coaching to support m-Health and using m-Health as a tool to facilitate coaching. The findings indicate that while m-Health technologies are effective in telecommunication and disease monitoring, patients showed a clear preference for physical interactions over digital communication. This suggests that although m-Health can be beneficial, it should not replace the human element in healthcare [9]. Prioritizing personalized approaches that combine human contact with technology could provide a more comprehensive solution for improving self-management in chronic care.

Particularly when it comes to managing chronic illnesses and encouraging physical activity (PA), mHealth has shown great promise as a tool for enhancing healthcare outcomes. Although findings for moderate-to-vigorous PA are still unclear, research indicates that smartphone-based interventions can dramatically enhance children's and adolescents' physical activity, especially in total PA and daily steps [10]. Self-efficacy and care quality are important factors in mHealth's support of health monitoring, communication, and decision-making in individuals with comorbidities [11]. Furthermore, the UTAUT paradigm has an impact on mHealth acceptance, especially in

underdeveloped nations like Bangladesh, where it has promise despite slower adoption rates [12].

A study using Activity Theory (AT) explores how mHealth applications are used in real-life contexts, emphasizing user adaptation and interaction beyond task-oriented technology use. It highlights how mHealth systems influence behavior, offering insights to improve design and implementation [13]. A review of security and privacy in m/uHealth systems categorized solutions, challenges, and trends. Most solutions since 2015 have been validated through prototypes or performance studies, with few undergoing real-life deployments, emphasizing the need for better evaluation methods [14]. The ROAMM-EHR platform integrates patient-generated health data (PGHD) from devices like smartwatches into electronic health records. Pilot testing showed improved post-surgical monitoring and clinical decisions, despite challenges like battery life and adoption barriers [15].

III. Spring Boot and React Architecture

A. Backend Services

The proposed healthcare analyser app utilizes the modular layered architecture based on the Spring Boot framework in order to store patient information and retrieve health information in an efficient manner. As shown in Fig. 1, the system is developed using Model-View-Controller pattern which presents a strong emphasis on well-defined distinct functional layers making the system easier to manage and easier to expand. The entry point for the HTTP requests concerning the health profiles of the patients, the Controller layer is used. It is a medium between the user interface and the service layer in charge of certain aspects of user interaction and return of results to the application. Once the request has been made, the controller processes the input, validates the input, and performs data processing functions before routing service requests to the business in charge of the details.

The Service layer handles the basic use case of fetching, editing, and saving patient related information. This lower layer combines a number of different repositories and utility components in order to perform the tasks. It is concerned with the data held in the repository layer and implements some of the business rules such as investigating blood pressure readings, evaluating life style practices and offering health promotion strategies.

The Repository layer will interact with the database by providing functionality to retrieve or modify the health profiles and other patient related data in the persistence layer. The repository system utilizes JPA (Java Persistence API) for data storage thus implementing CRUD (Create, Read, Update, and Delete) approaches. There is an implementation class which is in charge of creation of connections and executing the database commands through the entity manager. The JWT token manages user authentication and authorization, granting access to the application.

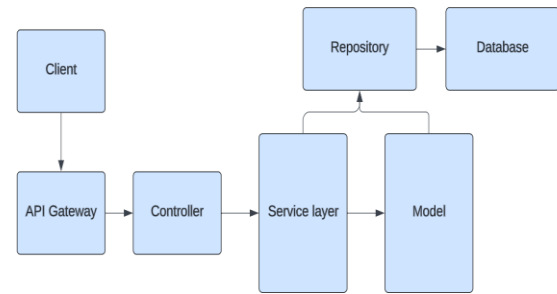


Fig. 1. Spring Boot Architecture

The Response utility standardizes API responses (e.g., in JSON/XML), while the Response Filter hides data based on the access token or request parameters. The app maintains a patient's health profile as a Value Object, including details like blood pressure, medications, meals, and exercises. This data model supports the retention and processing of patient data, enabling the app to deliver personalized health insights and recommendations.

B. Frontend Services

In the mHealth web app project, React plays a crucial role as the frontend framework. As seen in Fig. 2, the "Router" helps manage navigation between different views or pages, enabling single-page application (SPA) behaviour.

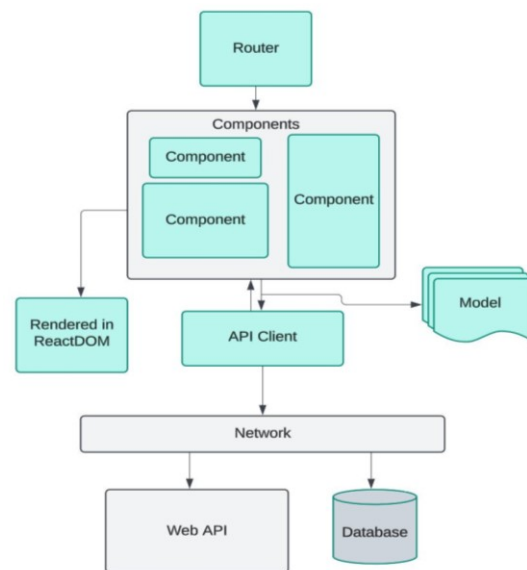


Fig. 2 : React Architecture

Each React component interacts with an API client to fetch data from a backend API, which is then rendered by ReactDOM to display changes efficiently on the browser. This process ensures that the web app can provide real-time updates, crucial for healthcare applications where responsiveness and performance are key. React's virtual DOM ensures that only the necessary components are re-rendered when the state changes, enhancing performance and

delivering a smooth user experience. React, as a framework, is designed for building user interfaces with a focus on efficiency and flexibility. It relies on a virtual DOM, which minimizes the number of direct changes to the real DOM, speeding up rendering and making the application more responsive. React follows a declarative approach, meaning developers describe how the UI should look at any given point, and React efficiently updates the view when the data changes. Its component-based structure also encourages reusability, allowing developers to maintain consistency across the application while reducing redundancy in the code.

C. Database

PostgreSQL was mainly used as the database to manipulate and store patient data related to the healthcare analysis application. This allows for easy handling of structured data associated with patient data and analytical outcomes. The application uploaded data as CSV, which was stored through PostgreSQL. The stored data was accessed for visualization, queries, and insight generation purposes. PostgreSQL integration ensures that the data storage is scalable and reliable, and it supports the requirements of an application for analytics and data management.

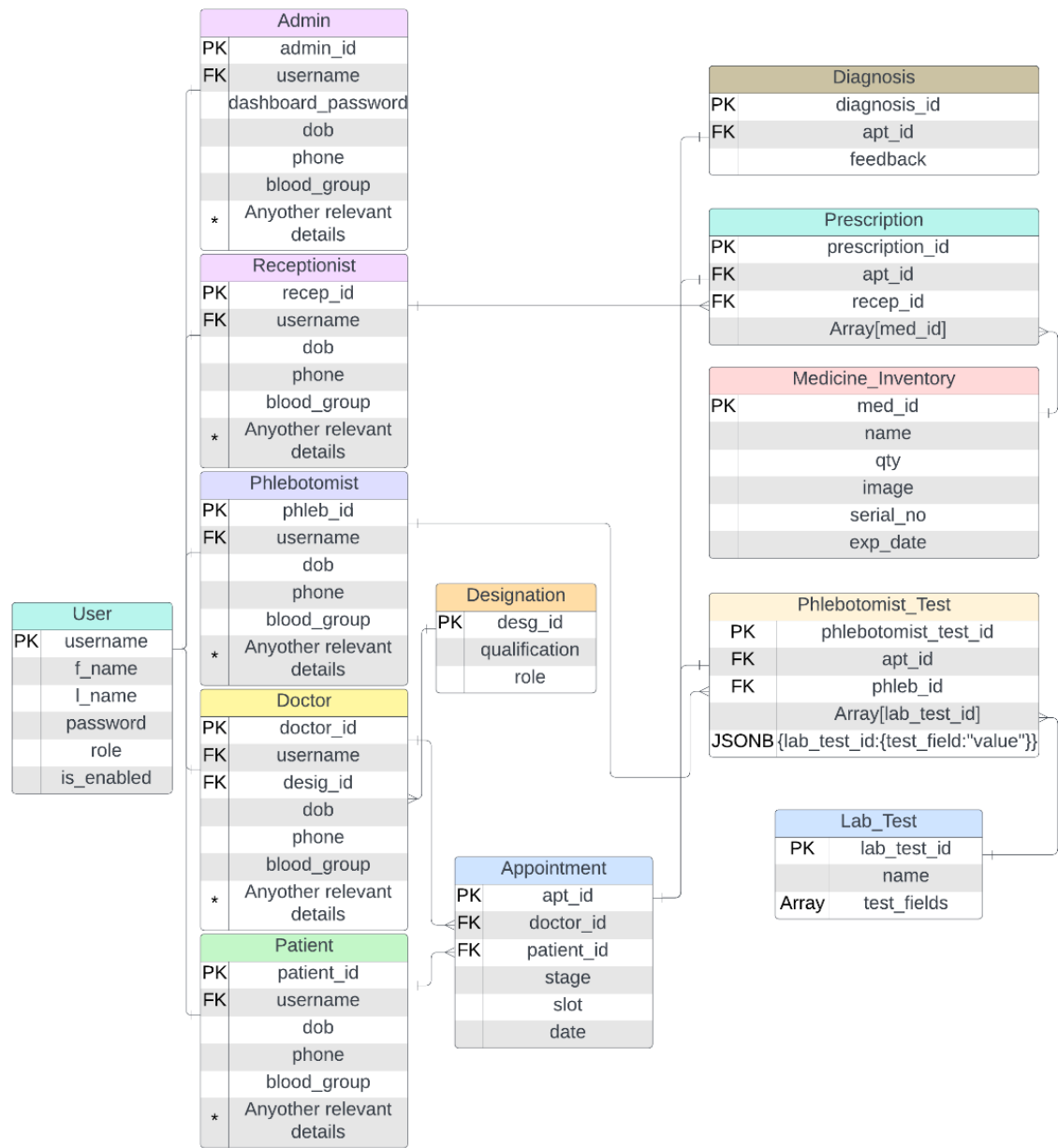


Fig. 3. Database Architecture

D. Services

The proposed methodology for our healthcare analysis application is about creating a platform that would be used to help users gain an efficient method of analysing and managing health-related data. This system is built with a structured, service-oriented architecture design, ensuring clear separation of concerns in the handling of data. As shown in Fig.3 this work is designed to streamline hospital management by organizing key functions into distinct user roles and processes. The system architecture supports various roles, including Admin, Receptionist, Doctor, Phlebotomist, and Patient, each represented by dedicated entities within the database schema. Admins manage the overall system, while Receptionists are responsible for coordinating appointments and prescriptions. Each Appointment entry links patients and doctors, enabling seamless scheduling and tracking of patient visits.

The Phlebotomist role is integrated into the management of laboratory tests, where the Phlebotomist_Test table records the outcomes of medical tests in a flexible JSONB format, ensuring efficient storage and retrieval of results. Doctors and other medical staff are categorized by their qualifications and roles through the Designation entity, maintaining a clear hierarchy and scope of responsibilities.

Additionally, the system incorporates a Medicine Inventory module that tracks the availability of medications, their quantities, expiration dates, and serial numbers, ensuring accurate stock management. The user authentication process is governed by the User entity, which ensures secure access control based on predefined roles. This modular design supports efficient hospital operations, enabling medical personnel and administrators to manage patient care, appointments, and inventory in a cohesive manner.

In this application, once a patient logs in, they can enter their symptoms and schedule an appointment with a doctor. The doctor assesses the symptoms during the consultation and may assign tests, which are managed through the system by phlebotomists. After the test results are available, the doctor provides a diagnosis and prescribes medication. Each stage—symptom entry, appointment, tests, diagnosis, and prescription—is efficiently tracked, ensuring smooth communication between patients and healthcare providers. This process helps streamline hospital workflows and improve patient care.

Therefore, this system ensures that all information is stored securely and can be easily accessed by authorized users at any point in the process. By automating these critical steps, the M-Health application reduces administrative burden and improves the overall patient experience

IV. Results and Discussion

In this Spring Boot based healthcare application, performance metrics reflect efficient resource utilization.

Fig. 4 shows low CPU usage, rarely exceeding 3%, which indicates that the application effectively handles operations without high processing demands. It also presents stable heap memory usage, with memory consumption remaining well within the 2048 MB threshold, demonstrating controlled memory allocation for consistent performance.

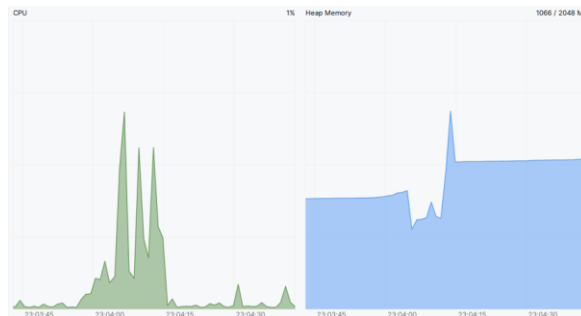


Fig. 4. CPU and Heap Memory Usage

Thread activity, as illustrated in Fig. 5, shows that the application uses 84 of 132 available threads, supporting efficient task management without overloading system resources. Non-heap memory is consistently utilized up to but not beyond its limit, ensuring reliable handling of background tasks and data caching.

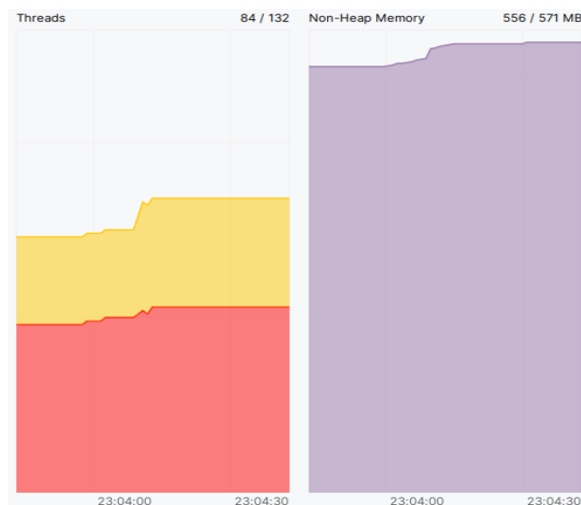


Fig. 5. Thread activity and Non-Heap Memory

Compared to MERN stack applications, which may experience higher resource consumption under heavy load, Spring Boot's multi-threading capabilities and built-in resource optimization make it better suited for demanding healthcare applications as seen in Fig. 6. This efficiency provides a scalable, adaptable framework ideal for handling high data volumes and supporting future enhancements like real-time monitoring and predictive analytics.

Feature	System A (MERN Stack)	System B (LAMP Stack)	Our System (Spring Boot)
End-to-End Service	Limited; needs integration with separate services	Partial; lacks smooth integration of services	Comprehensive; fully integrated for end-to-end service
Scalability	Moderate; can struggle with high traffic	Limited; may require significant resources for scaling	High; handles growth efficiently and supports complex data needs
Data Processing & Analytics	Basic; offers limited analytical tools and slower processing	Limited; data processing capabilities are restricted	Advanced; robust data handling and insightful analytics with fast processing

Fig. 6. Comparison Table

This table contrasts two alternative systems built on the MERN and LAMP stacks with our healthcare analytic system, which was created with Spring Boot. While System A (MERN) and System B (LAMP) need additional services to accomplish a full workflow, the Spring Boot system is unique for end-to-end services since it offers a smooth, completely integrated service model. In contrast to System A, which can lag under large loads, and System B, which has few scalability choices, Spring Boot provides great scalability, effectively handling increasing data demands. Finally, when it comes to analytics and data processing, our system offers strong data handling and produces informative visualisations fast, while the other systems just give rudimentary analytics and frequently have slower processing times. All things considered, the healthcare solution offered by the Spring Boot-based application is more scalable and integrated.

V. Conclusion

The M-Health application is a highly valuable asset for analyzing and managing healthcare data, offering healthcare professionals the ability to visualize and interpret vast amounts of patient information in a streamlined and efficient manner. By digitizing key processes, the system enhances patient care, allowing for better management of appointments, diagnoses, tests, and prescriptions. The platform simplifies data handling, making it easier for healthcare providers to generate meaningful insights that can improve medical practice and patient outcomes.

Looking ahead, the application has great potential for further advancement. Incorporating machine learning algorithms will enable predictive analytics, allowing healthcare professionals to detect early health risks and trends in patient data, leading to proactive care. Integrating real-time data

processing from wearable health devices will add continuous patient monitoring, offering more comprehensive health management. Furthermore, by implementing robust security features like user authentication and role-based access control, the system will adhere to data protection regulations, ensuring patient privacy and securing sensitive health information. These enhancements will make the M-Health application an even more powerful tool, extending its value across different aspects of healthcare delivery and decision-making.

REFERENCES

- [1] Calejari, L. P., Tortorella, G. L., & Fettermann, D. C. Getting connected to m-health technologies through a meta-analysis. *International Journal of Environmental Research and Public Health*, 20(5), 4369, pp. 8-14, 2023.
- [2] Istepanian, R.S.H. Mobile Health (m-Health) in Retrospect: The Known Unknowns. *Int. J. Environ. Res. Public Health*, 19, 3747, pp. 10-22, 2022.
- [3] Sezgin, E., & Yıldırım, S. Ö. (2014). A literature review on attitudes of health professionals towards health information systems: From e-health to m-health. *Procedia Technology*, 16, pp. 1317-1326, 2014.
- [4] Akbulut, A., Desouki, S., AbdelKhaliq, S. *et al.* Design and implementation of a deep learning-empowered m-Health application. *Multimed Tools Appl* 83, 35995–36011, 2024
- [5] Mishlanov, V., Chuchalin, A., Chereshev, V., Poberezhets, V., Vitacca, M., Nevzorova, V., Aisanov, Z., Vizel, A., Shubin, I., Nikitin, A., Zulkarneev, R., & Khovaeva, Y. (2019). Scope and new horizons for implementation of m-health/e-health services in pulmonology in 2019. *Monaldi Archives for Chest Disease*, 89(3), pp.6-8, 2019.
- [6] Sereno, M., Iniesta-Chamorro, J. M., Garrido-Rubiales, B., Gomez, E. J., & Casado Sáenz, E. M-health in lung cancer: A literature review. *SAGE Open Medicine*, 11, pp. 3-4, 2023
- [7] Eberle, C., Löhnert, M., & Stichling, S. Effectiveness of disease-specific mHealth apps in patients with diabetes mellitus: Scoping review. *JMIR Mhealth and Uhealth*, 9(2), e23477. 2021.
- [8] Obro, L. F., Heiselberg, K., Krogh, P. G., Handberg, C., Ammentorp, J., Pihl, G. T., & Osther, P. J. S. Combining mHealth and health-coaching for improving self-management in chronic care: A scoping review. *Patient Education and Counseling*, 104(4), 680-688. 2021.
- [9] Liu, F., Song, T., Yu, P., Deng, N., Guan, Y., Yang, Y., & Ma, Y. Efficacy of an mHealth app to support patients' self-

management of hypertension: Randomized controlled trial. *Journal of Medical Internet Research*, 25, e43809.

[10] He, Z., Wu, H., Yu, F., Fu, J., Sun, S., Huang, T., Wang, R., Chen, D., Zhao, G., & Quan, M. Effects of Smartphone-Based Interventions on Physical Activity in Children and Adolescents: Systematic Review and Meta-analysis. *JMIR MHealth and UHealth*, 9(2), e22601. 2021.

[11] Elkefi, S. Access and usage of mobile health (mHealth) for communication, health monitoring, and decision-making among patients with multiple chronic diseases (comorbidities). *IJSE Transactions on Healthcare Systems Engineering*, 14(3), 179–192. 2023.

[12] Alam, M. Z., Hoque, M. R., Hu, W., & Barua, Z. Factors influencing the adoption of mHealth services in a developing country: A patient-centric study. *International Journal of Information Management*, 50, pp. 128-14, 2020.

[13] Sowon, K., & Chigona, W. Conceptualising Healthcare-Seeking as an Activity to Explain Technology Use: A Case of M-health.1st Virtual Conference on Implications of Information and Digital Technologies for Development. 2021.

[14] Iwaya, L. H., Ahmad, A., & Babar, M. A. Security and Privacy for mHealth and uHealth Systems: A Systematic Mapping Study. *IEEE Access*, 8, 150081–150112, 2020.

[15] Sun, Heng, Sai Manoj Jalam, Havish Kodali, Subhash Nerella, Ruben D. Zapata, Nicole Gravina, Jessica Ray, Erik C. Schmidt, Todd M. Manini and Rashidi Parisa. “Enhancing EHR Systems with data from wearables: An end-to-end Solution for monitoring post-Surgical Symptoms in older adults.” 2024.